Experiment 1

Design and Verification of Half Adder, Full Adder & 4-Bit Adder on Logisim

Experiment No.	1	**Subject**	COA
Branch	CSE	**Name**	Anshul Modi
Roll No.	241210021		

1. Aim / Objective

To design and verify the operation of a **Half Adder**, **Full Adder**, and **4-bit Ripple Carry Adder** using basic logic gates on the Logisim simulation software and to validate the output against the expected truth table.

2. Theory

Half Adder: A half adder is a combinational circuit that adds two single-bit binary numbers (A and B) and produces two outputs: **Sum (S)** and **Carry (C)**. The Sum is obtained using an XOR gate and the Carry using an AND gate.

$$S = A \oplus B \qquad C = A \cdot B$$

Full Adder: A full adder extends the half adder by accepting a **carry input (Cin)** in addition to two input bits. It produces a Sum and a Carry output. A full adder can be built using two half adders and an OR gate.

$$S = A \oplus B \oplus Cin \qquad Cout = (A \cdot B) + (Cin \cdot (A \oplus B))$$

4-Bit Ripple Carry Adder: Four full adders are cascaded such that the carry output of each stage is fed as the carry input to the next stage. The least significant bit stage has Cin = 0. The result is a 4-bit sum with a final carry-out.

3. Truth Tables

3.1 Half Adder Truth Table

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

3.2 Full Adder Truth Table

A	B	Cin	Sum (S)	Carry (Cout)
0	0	0	0	0

0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

4. Logic Expressions (Boolean)

Circuit	Expression
Half Adder – Sum	$S = A \oplus B$
Half Adder – Carry	$C = A \cdot B$
Full Adder – Sum	$S = A \oplus B \oplus Cin$
Full Adder – Carry	$Cout = AB + Cin(A \oplus B)$
4-bit Adder	Four FA stages; Cin of FA0 = 0; Cout of FAn → Cin of FA(n+1)

5. Procedure

1. Open Logisim and create a new project.
2. Place input pins (A, B for half adder; A, B, Cin for full adder) from the wiring library.
3. Add an XOR gate for the Sum output and an AND gate for the Carry output to build the Half Adder.
4. Verify the Half Adder output against the truth table by toggling input values.
5. Build the Full Adder by connecting two Half Adders with an additional OR gate for Cout.
6. Verify the Full Adder outputs for all 8 input combinations.
7. Cascade four Full Adder modules, connecting Cout of each to Cin of the next, to form the 4-bit Adder.
8. Test the 4-bit adder with multiple binary operand pairs and verify the sum and carry-out.
9. Take screenshots of each verified circuit for the record.

6. Logisim Circuit Screenshots

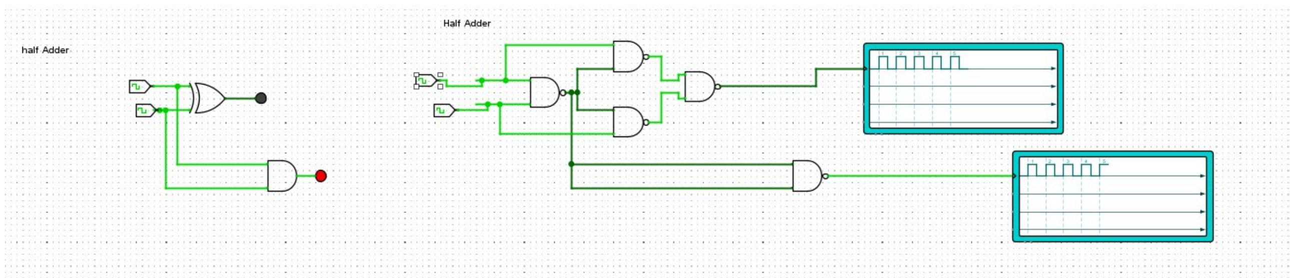

Fig. 1 – Half Adder circuit on Logisim.

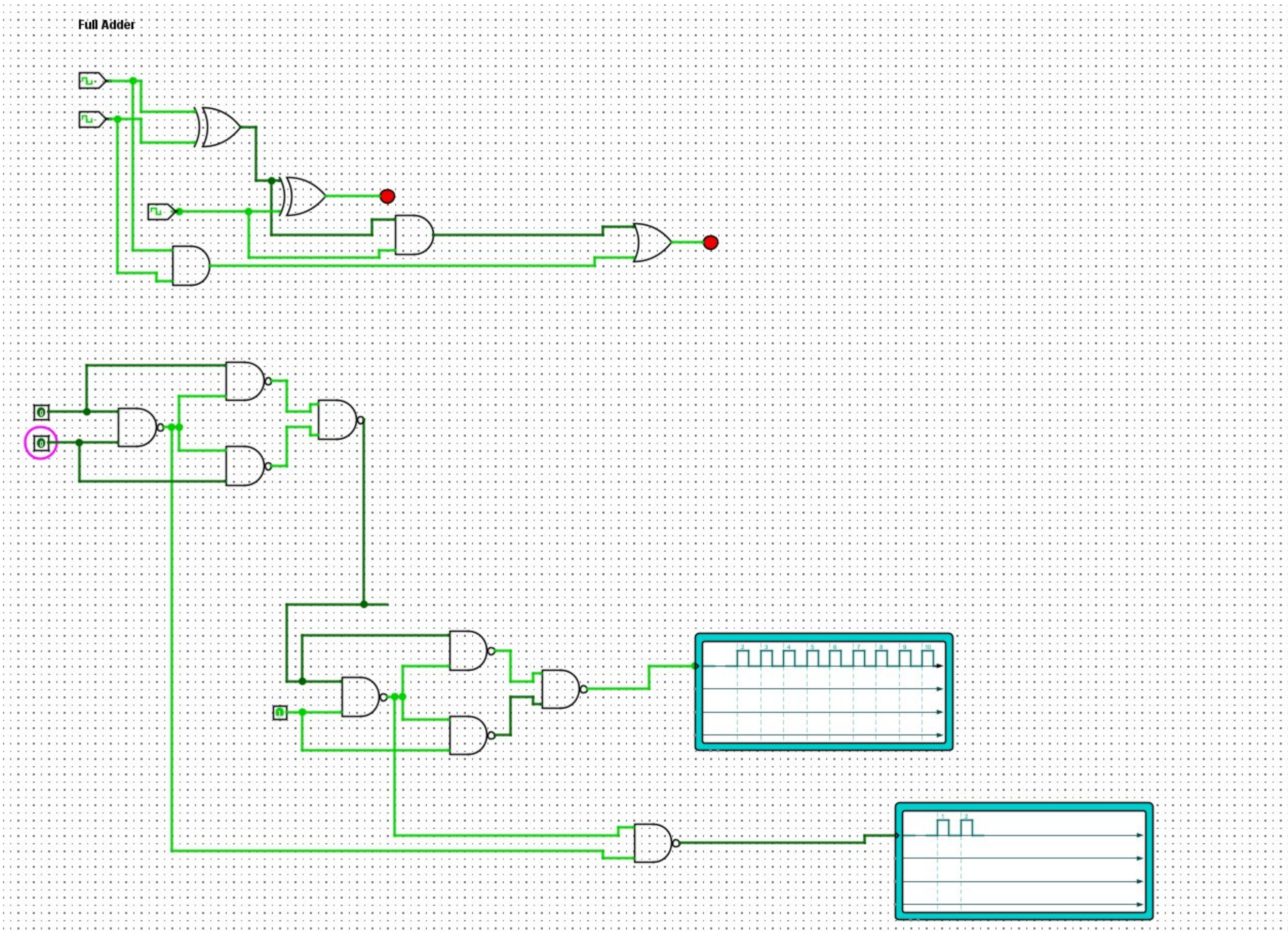

Fig. 2 – Full Adder circuit on Logisim.

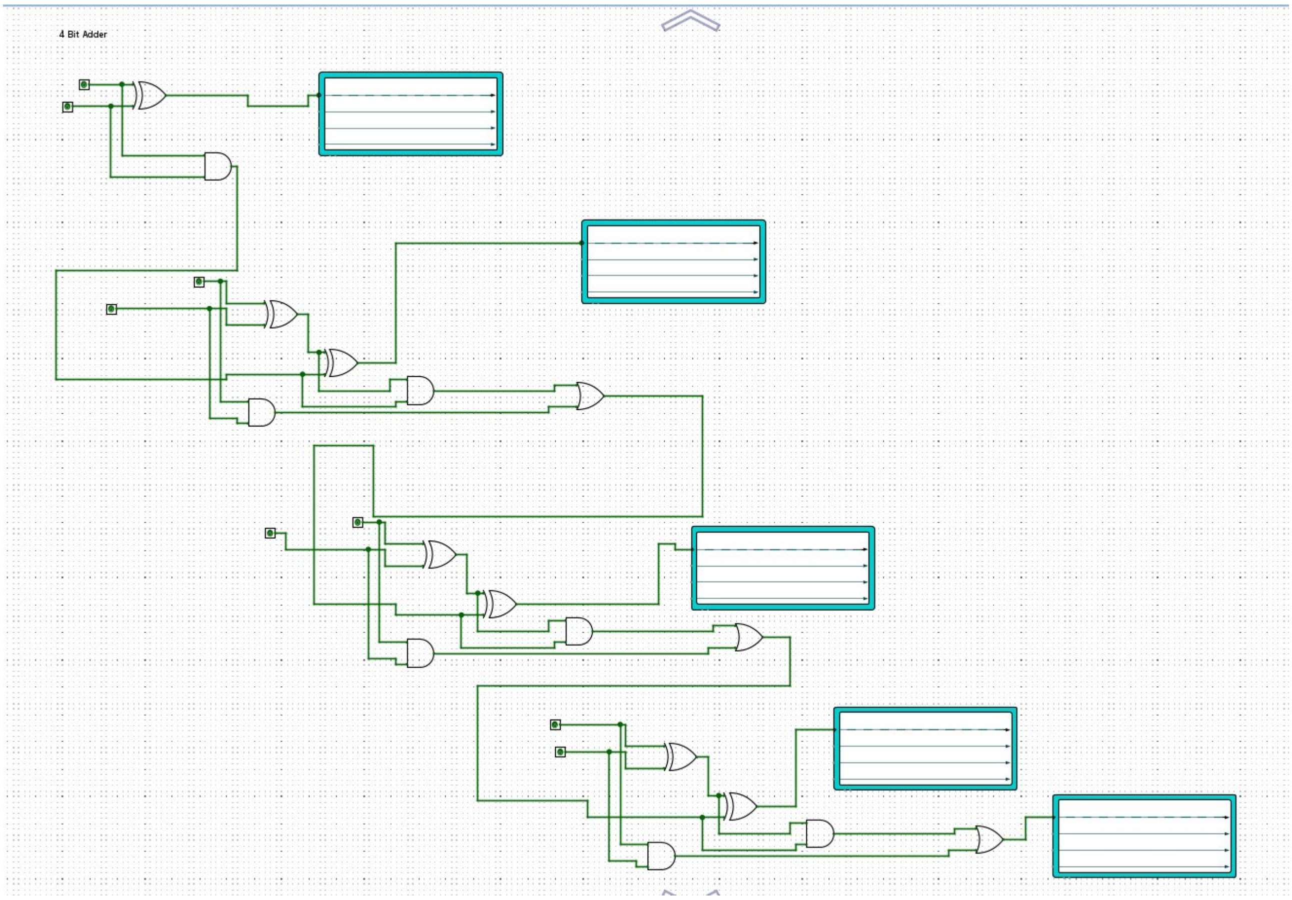

Fig. 3 – 4-Bit Ripple Carry Adder circuit on Logisim.

7. Observations

All simulated outputs in Logisim matched the expected truth table values for the half adder, full adder, and 4-bit adder. The carry propagation in the 4-bit adder was observed to be sequential (ripple), confirming correct cascading of full adder stages.

8. Result

The half adder, full adder, and 4-bit ripple carry adder were successfully designed and verified on Logisim. The circuit outputs matched the theoretical truth tables, thereby confirming the correctness of the design.